

knowledge, students should have opportunities to engage with fabrication plants, packaging lines, validation laboratories, and real manufacturing processes. This approach would help develop industry-ready talent and address the skills gap as manufacturing expands.

The government should prioritize the development of shared national infrastructure for silicon validation, advanced packaging, and reliability testing. Providing access to centralized, high-quality facilities would enable startups and MSMEs to transition from design to production-ready silicon without the need for each company to independently invest in expensive infrastructure.

What semiconductor roles are hardest to hire for in India today?

Jaya: One of the toughest talent

gaps in semiconductors today is the shortage of engineers with deep, end-to-end experience across the chip design flow. While India has strong execution capability, highly experienced architects and technical experts who can define new architectures, design flows and products from first principles are far harder to find. Unlike more mature ecosystems that have built silicon for decades, India has a smaller pool of professionals with long, multi-cycle silicon experience, and as chip complexity increases, this kind of deep technical and architectural talent is in especially high demand.

Alongside this, new skill sets are gaining importance. The ability to apply AI and machine learning directly to semiconductor design whether for automation, optimization or early defect detection is becoming central

to modern chip development. There is also rising demand for secure hardware design expertise, as security at the silicon level increasingly becomes a baseline requirement built-in rather than an add-on.

How should India realistically measure its success in semiconductor ecosystem over the next five years?

Jaya: India's semiconductor success over the next five years can be measured on how many products are actually designed and built in India for the Indian market. That means seeing local companies develop products that are used within the country. The country should focus on meeting domestic needs and building confidence in made-in-India semiconductor products. **d**

